Microeconomic Analysis

Static Optimisation

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Static Optimisation

Topics

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IV. Concave and Convex Problems

Global vs Local Optima

Consider the general problem

$$\text{maximise } f(\mathbf{x}) \text{ subject to } \mathbf{x} \in S$$

So far we have used the *necessary* FOC and the *sufficient* SOC to find local optima.

When can we guarantee that a local optimum is also a global one?

Concave and Convex Functions

Definition

Consider a convex set $S \subseteq \mathbb{R}^n$. A real-valued function $f : S \to \mathbb{R}$ is concave (convex) if for any $\mathbf{x}, \mathbf{y} \in S$, and any $t \in (0,1)$

$$f(t\mathbf{x} + (1-t)\mathbf{y}) \geq (\leq) tf(\mathbf{x}) + (1-t)f(\mathbf{y})$$

The properties are strict if the inequalities are strict.

Concave and Convex Functions

Lemma

If $f : S \to \mathbb{R}$ is C^1 then, f is concave (convex) if and only if:

$$f(\mathbf{y}) - f(\mathbf{x}) \leq (\geq) Df(\mathbf{x})(\mathbf{y} - \mathbf{x}) \quad \text{for all } \mathbf{x}, \mathbf{y} \in S$$

In particular, if $\mathbf{x}_0$ is such that $Df(\mathbf{x}_0) = 0$, then:

- if f is concave $\longrightarrow$ $\mathbf{x}_0$ is a global maximum
- if f is convex $\longrightarrow$ $\mathbf{x}_0$ is a global minimum

Concave and Convex Functions

If the function $f : S \to \mathbb{R}$ is C^2, we already saw another way of checking whether the function is concave/ convex.

Lemma

If $f : S \to \mathbb{R}$ is C^2 then,

- $D^2 f$ is negative semi-definite for all $\mathbf{x} \in S \iff f$ is concave on S.
- $D^2 f$ is negative definite for all $\mathbf{x} \in S \implies f$ is strictly concave on S.
- $D^2 f$ is positive semi-definite for all $\mathbf{x} \in S \iff f$ is convex on S.
- $D^2 f$ is positive definite for all $\mathbf{x} \in S \implies f$ is strictly convex on S.

Concave and Convex Functions

P: maximise $f(\mathbf{x})$ subject to $\mathbf{x} \in S$

Global Optima

Let f be (strictly) concave/convex in S.

- If a stationary point exists, it is a (unique) global maximum/minimum.
- If a local maximum/minimum exists, it is a (unique) global maximum/minimum.

Interior critical (stationary) points are automatically global optima (FOC are sufficient).

However, for constrained candidates, we still need to check SOC to see whether they are local constrained optima.

When can we guarantee that the FOC are *sufficient* for a global optima?

Lagrangian Sufficiency

P: maximise $f(\mathbf{x})$ subject to $\mathbf{g}(\mathbf{x}) \leq \mathbf{b}$

Lagrangian Sufficiency

If there exist $\mathbf{x}^*$ and $\boldsymbol{\lambda}^* \geq 0$ such that

$$L(\mathbf{x}^*, \boldsymbol{\lambda}^*) \geq L(\mathbf{x}, \boldsymbol{\lambda}^*) \quad \text{for all } \mathbf{x} \in \mathbb{R}^n \tag{1}$$

$$\boldsymbol{\lambda}^* \left(\mathbf{g}\left(\mathbf{x}^*\right) - \mathbf{b}\right) = 0 \tag{2}$$

and $\mathbf{x}^*$ is feasible ($\mathbf{g}(\mathbf{x}^*) \leq \mathbf{b}$), then $\mathbf{x}^*$ is optimal for P.

Proof: Consider any $\mathbf{x}$ which is feasible for P (i.e., $\mathbf{g}(\mathbf{x}) \leq \mathbf{b}$),

$$f(\mathbf{x}^*) = L(\mathbf{x}^*, \boldsymbol{\lambda}^*) \geq L(\mathbf{x}, \boldsymbol{\lambda}^*) \geq f(\mathbf{x})$$

The equality follows from (2) and the second inequality from the fact that $\mathbf{x}$ is feasible and $\boldsymbol{\lambda}^* \geq 0$. Hence $\mathbf{x}^*$ is optimal.

Lagrangian Sufficiency

Comments:

- The Lagrangian sufficiency says that **if** we are able to find a feasible $\mathbf{x}^*$ and a $\boldsymbol{\lambda}^* \geq 0$ with $\boldsymbol{\lambda}^* \left(\mathbf{g}\left(\mathbf{x}^*\right) - \mathbf{b}\right) = 0$, such that $\mathbf{x}^*$ maximises the Lagrangian $L(\mathbf{x}, \boldsymbol{\lambda}^*)$, then $\mathbf{x}^*$ is a global optimum.

- We might not be able to find such a $\boldsymbol{\lambda}^*$ and $\mathbf{x}^*$...

- Note that the Lagrangian sufficiency result does not make any concavity or convexity assumption so can be used to prove global optimality even when these assumptions do not hold.

- Note that we did not require any Constraint Qualification either.

Concave and Convex Problems

Definition: Concave Problem

Consider the problem:

$$\text{P: maximise } f(\mathbf{x}) \text{ subject to } \mathbf{g}(\mathbf{x}) \leq \mathbf{b}$$

P is a Concave Problem if f is concave and each g_j is convex.

Definition: Convex Problem

Analogously, the problem:

$$\text{P': minimise } f(\mathbf{x}) \text{ subject to } \mathbf{g}(\mathbf{x}) \geq \mathbf{b}$$

is a Convex Problem if f is convex and each g_j is concave.

Concave and Convex Problems

Sufficient FOC

Let P be a concave (convex) problem, then the Kuhn-Tucker first order conditions are sufficient for a global maximum (minimum).

Note that the CQ is not required for this result.

Proof: If $\boldsymbol{\lambda} \geq 0$ then since f is concave and the g_j convex, the Lagrangian

$$L(\mathbf{x}, \boldsymbol{\lambda}) = f(\mathbf{x}) - \boldsymbol{\lambda}(\mathbf{g}(\mathbf{x}) - \mathbf{b})$$

is a concave function of $\mathbf{x}$.
Let $(\mathbf{x}^*, \boldsymbol{\lambda}^*)$ satisfy the Kuhn-Tucker first-order conditions. These imply that $\mathbf{x}^*$ is a stationary point of $L(\mathbf{x}, \boldsymbol{\lambda}^*)$ with respect to $\mathbf{x}$. Since L is concave this implies that $\mathbf{x}^*$ maximises $L(\mathbf{x}, \boldsymbol{\lambda}^*)$ globally with respect to $\mathbf{x}$.

By the Lagrangian sufficiency, $\mathbf{x}^*$ is optimal for P.

Optimisation: Concavity and Convexity

Comments:

- Without the concavity assumptions, a solution to the K-T conditions needs not maximise $L(\mathbf{x}, \boldsymbol{\lambda}^*)$, and the Lagrangian sufficiency does not apply.
- These are sufficient conditions. It could be that the K-T conditions don't have a solution... However, if f is concave and g_j is convex, then if in addition $\mathbf{g}(\mathbf{x}) << \mathbf{b}$ for some feasible $\mathbf{x}$, the global maximum will also maximise the Lagrangian.

Lagrangian Necessity

Let f be concave and let each g_j be convex.

Suppose, in addition, $\mathbf{g}(\mathbf{x}) << \mathbf{b}$ for some feasible $\mathbf{x}$.

If $\mathbf{x}^*$ is a global maximum there exists a vector $\boldsymbol{\lambda}^* \geq 0$, with

$$\boldsymbol{\lambda}^* \left(\mathbf{b} - \mathbf{g}\left(\mathbf{x}^*\right)\right) = \mathbf{0}$$

such that $\mathbf{x}^*$ maximises the Lagrangian

$$L = f(\mathbf{x}) + \boldsymbol{\lambda}^* \left(\mathbf{b} - \mathbf{g}\left(\mathbf{x}\right)\right)$$

Optimisation: Quasi-Concavity and Quasi-Convexity

Can we relax the concavity assumptions slightly and still have sufficient FOC?

Suppose that the domain S is convex.

Definition

- f is quasi-concave if the upper contour set $\{\mathbf{x} : f(\mathbf{x}) \geq c\}$ is a convex set for any real c;
- f is quasi-convex if the lower contour set $\{\mathbf{x} : f(\mathbf{x}) \leq c\}$ is a convex set for any real c.

The properties are strict if the upper (lower) contour sets are strictly convex.

Optimisation: Quasi-Concavity and Quasi-Convexity

Other characterisations of quasi-concave functions:

Lemma

A function f is quasi-concave if and only if

$$f(t\mathbf{x} + (1-t)\mathbf{y}) \geq \min\{f(\mathbf{x}), f(\mathbf{y})\} \; \forall t \in (0,1)$$

A function f is quasi-convex if and only if

$$f(t\mathbf{x} + (1-t)\mathbf{y}) \leq \max\{f(\mathbf{x}), f(\mathbf{y})\} \; \forall t \in (0,1)$$

The properties are strict if the inequalities are strict for $\mathbf{x} \neq \mathbf{y}$.

Optimisation: Quasi-Concavity and Quasi-Convexity

Lemma

If f a C^1 function,

f is quasi-concave if and only if

$$f(\mathbf{y}) \geq f(\mathbf{x}) \;\Rightarrow\; Df(\mathbf{x})(\mathbf{y}-\mathbf{x}) \geq 0$$

f is quasi-convex if and only if

$$f(\mathbf{y}) \leq f(\mathbf{x}) \;\Rightarrow\; Df(\mathbf{x})(\mathbf{y}-\mathbf{x}) \leq 0$$

The conditions for strict quasi-concavity are as follows:

If for all $\mathbf{x} \neq \mathbf{y}$, $Df(\mathbf{x})(\mathbf{y}-\mathbf{x}) > 0$ whenever $f(\mathbf{y}) \geq f(\mathbf{x})$, then f is strictly quasi-concave.

If f is strictly quasi-concave and $Df(\mathbf{x}) \neq 0$, then $Df(\mathbf{x})(\mathbf{y}-\mathbf{x}) > 0$ whenever $f(\mathbf{y}) \geq f(\mathbf{x})$.

Optimisation: Quasi-Concavity and Quasi-Convexity

Properties:

- f is quasi-concave $\Leftrightarrow -f$ is quasi-convex.
- f is concave $\implies f$ is quasi-concave.
- f is convex $\implies f$ is quasi-convex.
- Any (strictly) monotonic function of *one variable* is both (strictly) quasi-concave and (strictly) quasi-convex.

Example:

- $\ln(x)$ is concave, so also quasi-concave; increasing, so also quasi-convex (but not convex).
- $\ln(x) + \ln(y)$ is concave, so also quasi-concave; but not quasi-convex, so also not convex.

Optimisation: Quasi-Concavity and Quasi-Convexity

$$\text{maximise } f(\mathbf{x}) \text{ subject to } \mathbf{x} \in S$$

Global Optima

Let f be strictly quasi-concave.

If a local maximum exists, it is a unique global maximum.

Consider the constrained problem:

$$\text{maximise } f(\mathbf{x}) \text{ subject to } \mathbf{g}(\mathbf{x}) \leq \mathbf{b}$$

Sufficient FOC

Let f be quasi-concave and let each g_j be quasi-convex.

The Kuhn-Tucker first order conditions are sufficient for a global maximum (at $\mathbf{x}^*$) provided at least one of the following holds:

(1) f is concave (2) $Df(\mathbf{x}^*) \neq 0$

IV. Envelope Theorems

Envelope Theorems

How does the maximised value of an objective function (for example profits or utility) change as parameters (for example prices) change?

Suppose the objective function and the constraints depend on a set of parameters $\mathbf{a}$:

$$\max_{\mathbf{x}} f(\mathbf{x};\mathbf{a}) \quad \text{subject to} \quad \mathbf{g}(\mathbf{x};\mathbf{a}) \leq \mathbf{b}$$

The maximum value function (of the parameters) is

$$M(\mathbf{a}) = \{\max_{\mathbf{x}} f(\mathbf{x};\mathbf{a}) \,|\, \mathbf{g}(\mathbf{x};\mathbf{a}) \leq \mathbf{b}\}$$

If $\mathbf{x}^*(\mathbf{a})$ is the optimal value of $\mathbf{x}$ as a function of $\mathbf{a}$ then

$$M(\mathbf{a}) = f(\mathbf{x}^*(\mathbf{a}), \mathbf{a})$$

How does $M(\mathbf{a}) = f(\mathbf{x}^*(\mathbf{a}); \mathbf{a})$ change if a_i changes?

Envelope Theorems for Unconstrained Problems

$$M(\mathbf{a}) = f(\mathbf{x}^*(\mathbf{a}); \mathbf{a})$$

If there are no constraints, $\mathbf{x}^*(\mathbf{a})$ satisfies the FOCs:

$$\frac{\partial f}{\partial x_i}(\mathbf{x}^*(\mathbf{a}); \mathbf{a}) = 0 \quad \text{for } i = 1, ..., n$$

If the conditions for the IFT hold:

- f_{x_i} is C^1 (guaranteed if f is C^2)
- $\frac{\partial^2 f}{\partial x_i \partial a}(\mathbf{x}^*(\mathbf{a}); \mathbf{a}) \neq 0$

then $\mathbf{x}^*$ is a C^1 function of a.

Envelope Theorems for Unconstrained Problems

$$M(\mathbf{a}) = f(\mathbf{x}^*(\mathbf{a}); \mathbf{a})$$

Envelope Theorem

$$\frac{\partial M(\mathbf{a})}{\partial a_j} = \frac{\partial f(\mathbf{x}^*(\mathbf{a}); \mathbf{a})}{\partial a_j}$$

Proof: By the chain rule,

$$\frac{\partial M(\mathbf{a})}{\partial a_j} = \sum_i \frac{\partial f(\mathbf{x}^*(\mathbf{a}); \mathbf{a})}{\partial x_i} \frac{\partial x_i^*(\mathbf{a})}{\partial a_j} + \frac{\partial f(\mathbf{x}^*(\mathbf{a}); \mathbf{a})}{\partial a_j}$$

But $\frac{\partial f}{\partial x_i}(\mathbf{x}^*(\mathbf{a}); \mathbf{a}) = 0$.

Only the direct effect of a change of a_j on f matters. (And not the change through the change in $\mathbf{x}^*(\mathbf{a})$).

Envelope Theorems for Unconstrained Problems

Example:

Consider $f(x; a, b) = 4 + 2b\ln(x) - ax^2$, with $a \neq 0$.

- FOC:

$$f_x(x^*; a, b) = 0, \quad \text{i.e.} \quad \frac{2b}{x^*} - 2ax^* = 0 \quad \text{or} \quad (x^*(a, b))^2 = \frac{b}{a}.$$

- ET:

$$\frac{\partial M(a, b)}{\partial a} = \frac{\partial f(x^*(a); a)}{\partial a} = -x^*(a, b)^2 = -b/a$$

- Of course, you could have done it by substitution:

  $M(a, b) = f(x^*(a, b); a, b) = 4 + 2b\ln\sqrt{\frac{b}{a}} - b$, hence

$$\frac{\partial M}{\partial a} = -2b\sqrt{a}\frac{1}{2}a^{-3/2} = -\frac{b}{a}$$

Envelope Theorems for Constrained Problems

Assume that $\mathbf{x}^*(\mathbf{a})$ satisfies the Kuhn-Tucker first order conditions and that the conditions of the IFT hold (so that $\mathbf{x}^*(\mathbf{a})$ and $\boldsymbol{\lambda}^*(\mathbf{a})$ are C^1 functions).

Construct the Lagrangian:

$$L(\mathbf{x}; \boldsymbol{\lambda}; \mathbf{a}) = f(\mathbf{x}; \mathbf{a}) - \boldsymbol{\lambda}(\mathbf{g}(\mathbf{x}; \mathbf{a}) - \mathbf{b})$$

Envelope Theorem

$$\frac{\partial M(\mathbf{a})}{\partial a_i} = \frac{\partial L(\mathbf{x}^*(\mathbf{a}); \boldsymbol{\lambda}^*(\mathbf{a}); \mathbf{a})}{\partial a_i}$$

Proof: (For the simplicity: 1 variable, 1 parameter and 1 constraint $g(x; a) \leq 0$)

The Lagrangian is: $L(x; \lambda; a) = f(x; a) - \lambda g(x; a)$

By complementary slackness $\lambda^*(a)g(x^*(a), a) = 0$, hence

$$M(a) \equiv f(x^*(a); a) = L(x^*(a), \lambda^*(a), a)$$

Envelope Theorems for Constrained Problems

Hence

$$\frac{\partial M}{\partial a} = \frac{\partial L}{\partial x}\frac{\partial x^*}{\partial a} + \frac{\partial L}{\partial \lambda}\frac{\partial \lambda^*}{\partial a} + \frac{\partial L}{\partial a}$$

Now $\frac{\partial L}{\partial x} = 0$ at $x = x^*(a)$ and $\lambda = \lambda^*(a)$ by the first-order condition for x.

On the other hand

$$\frac{\partial L}{\partial \lambda} = -g(x^*(a), a)$$

so the second term becomes: $-g(x^*(a), a)\frac{\partial \lambda^*}{\partial a}$.

Differentiating the complementary slackness constraint $(\lambda^*(a)g(x^*(a), a) = 0)$ and re-arranging:

$$g(x^*(a), a)\frac{\partial \lambda^*}{\partial a} = -\lambda^*(a)\left[\frac{\partial g}{\partial x}\frac{\partial x^*}{\partial a} + \frac{\partial g}{\partial a}\right]$$

By complementary slackness either $g(x^*(a), a) = 0$ or $\lambda^*(a) = 0$.

In either case

$$g(x^*(a), a)\frac{\partial \lambda^*}{\partial a} = 0 \quad \text{hence,} \quad \frac{\partial M}{\partial a} = \frac{\partial L}{\partial a}$$

Envelope Theorems for Constrained Problems

Interpretation of Lagrange Multipliers:

We already said that λ_j represents the effect on the objective function of relaxing the constraint $g_j(\mathbf{x}) \leq b_j$.

This is an special case of the ET:

$$L(\mathbf{x}, \boldsymbol{\lambda}; \mathbf{b}) = f(\mathbf{x}) - \boldsymbol{\lambda}(\mathbf{g}(\mathbf{x}) - \mathbf{b})$$

$$M(\mathbf{b}) = f(\mathbf{x}^*(\mathbf{b})) \equiv L(\mathbf{x}^*(\mathbf{b}), \boldsymbol{\lambda}^*(\mathbf{b}); \mathbf{b})$$

$$\frac{\partial M}{\partial b_j} = \frac{\partial L}{\partial b_j} = \lambda_j^*$$

To Sum Up: Static Optimisation

In these lectures we have looked at necessary and sufficient conditions for optima in:

- Unconstrained problems,
- Constrained problems with:
  - Equality constraints,
  - Inequality constraints,
  - Mixed problems.

For *concave* problems the FOC are not only necessary but also sufficient for global optima.

We have covered the Envelope Theorem for static optimisation which allows to do comparative statics of maximum value functions.